

Cyclospora cayetanensis

**Background document for the
WHO Guidelines for drinking-water quality and the
WHO Guidelines on sanitation and health**

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Highlights

- There is moderate evidence linking transmission of *Cyclospora cayetanensis* with unsafe drinking-water, inadequate sanitation, and poor hygiene. Drinking-water outbreaks have been documented although not on the same scale as other enteric protozoan pathogens.
- *Cyclospora cayetanensis* is the only *Cyclospora* species known to infect humans. Humans are the only known host.
- Transmission is primarily by the faecal-oral route. Food is the dominant source, but water has been identified as a possible source. Person-to-person transmission is unlikely as excreted oocysts are not immediately infective.
- *Cyclospora cayetanensis* causes watery diarrhoea, abdominal cramping, weight loss, anorexia, myalgia, fatigue, and occasionally vomiting and fever. The disease may be more severe in immunocompromised individuals.
- Infections occur worldwide but are much more common in tropical areas. In endemic areas infection and illness are most common in children less than 5 years of age. Nevertheless, the childhood diarrhoeal disease burden attributable to *C. cayetanensis* in low-income, high-burden settings is low in comparison to other gastrointestinal pathogens. Initial infection may provide protection against illness and asymptomatic infection can be very common in children in endemic areas. Infection is associated with travel to endemic areas or consumption of food from endemic areas.
- It has been suggested that infectivity is high, but the infective dose has not been determined.
- *Cyclospora* oocysts can be detected in environmental samples using microscopy or molecular methods, but standard methods are not available.
- Limited numbers of studies have detected *Cyclospora* oocysts in wastewater, surface water, groundwater, drinking-water and on crops. They can persist in water for long periods of time.
- Systematic approaches that identify and control faecal contamination along the whole sanitation service and drinking-water supply chains, are needed to manage microbial risks, including from *C. cayetanensis*.
- There are limited data on removal of *Cyclospora* by wastewater and drinking-water treatment processes. Removal is likely based on physical removal (e.g. size exclusion). Limited evidence suggests oocysts are highly resistant to chemical disinfectants.
- Due to the long environmental persistence of *Cyclospora* oocysts and their probable high resistance to chemical disinfectants, the absence of *E. coli* (or alternatively, thermotolerant coliforms) is not a suitable indicator of the absence of *C. cayetanensis* in drinking-water. However, the presence of *E. coli* is a suitable indicator of the potential presence of *C. cayetanensis*.

1 General description

Cyclospora cayetanensis is a single-celled, obligate, intracellular, coccidian protozoan parasite (Shields & Olson, 2003). It produces oocysts which are spheroidal in shape and between 8 µm to 10 µm in diameter with two-layered walls. Excreted oocysts must undergo sporulation in the environment to become infectious to humans. The environmental factors that initiate sporulation and development of infectivity are not fully known. Infective oocysts contain two sporocysts, each with 62 nm-thick walls surrounding a plasma membrane containing two sporozoites. Nineteen species of *Cyclospora* have been described, of which *C. cayetanensis* is the only species in the genus known to infect humans. Humans are the only known host of *C. cayetanensis* (Chacín-Bonilla, 2017; Almeria, 2019).

2 Human health effects

Infection by *C. cayetanensis* (cyclosporiasis) is characterized by watery diarrhoea, abdominal cramping, weight loss, anorexia, myalgia, fatigue, and occasionally vomiting and fever. Serious sequelae such as Guillain-Barré syndrome and Reiter's syndrome have been reported but not frequently enough to give an estimate of the risk (Mathison & Pritt, 2021).

The severity and course of infection varies from patient to patient, depending on immune status (Giangaspero & Gasser, 2019). In otherwise healthy individuals, symptoms are usually self-limiting but relapsing illness can last weeks or months. Increased susceptibility, prolonged diarrhoea, high rates of reoccurrence and severe intestinal injury have been observed in immunocompromised patients, particularly those with HIV or HIV/TB co-infections (Shields & Olson, 2003; Heymann, 2015; Ramezanzadeh et al., 2022).

Asymptomatic infection is common in endemic areas. Early and persistent exposure is associated with decreased likelihood of symptomatic infection (Chacín-Bonilla, 2010; Bhandari et al., 2015).

Trimethoprim-sulfamethoxazole is the drug of choice for managing cyclosporiasis (Chacín-Bonilla, 2017).

3 Epidemiology

Cyclospora cayetanensis has a global distribution with predominance in low-income, tropical and sub-tropical countries where it is often endemic (Ortega et al., 1998; Chacín-Bonilla 2010). Community-based studies in endemic settings have found widely varying infection prevalence rates ranging from 0% to over 40% (Giangaspero & Gasser, 2019). The age distribution of infection in the community also varies. Studies from some countries report the highest prevalence in children aged 2 to 4 years (Chacín-Bonilla, 2017). Others have found a higher prevalence in older, school-aged children, perhaps due to different exposure patterns associated with age in different areas (Chacín-Bonilla, 2010). Studies conducted in health-care settings among those with clinical manifestations show that the highest risk of infection with diarrhoea occurs in the first 5 years of life (Chacín-Bonilla, 2017). A re-analysis of the Global Enteric Multicentre Study (GEMS) found that the moderate to severe childhood diarrhoeal disease

burden attributable to *C. cayetanensis* in low-income, high-burden settings was low in comparison to other gastrointestinal pathogens (Liu et al., 2016). However, attribution in endemic settings is often complicated due to the high proportion of asymptomatic cases (Chacín-Bonilla, 2010).

In several endemic countries, the prevalence of *C. cayetanensis* has a marked seasonality. It often peaks in the warm and rainy seasons, although sometimes it occurs in the dry seasons (Almeria et al., 2019). This variability suggests that the life cycle of *C. cayetanensis* in the environment is likely to be influenced by several factors, such as rainfall, temperature, and humidity.

In non-endemic settings, infections are observed in the general population regardless of age and immune status and are almost always symptomatic (Gajadhar et al., 2015). Illness is linked mostly to travel to endemic areas or consumption of food from endemic areas (Herwaldt & Ackers, 1997; Alfano-Sobsey et al., 2004; Chacín-Bonilla, 2010; CDC, 2013).

4 Transmission

Cyclospora cayetanensis is transmitted by the faecal-oral route. The primary source of transmission in low-income countries appears to be food. Raw fruits and vegetables are most frequently identified as the source of transmission, but contact with contaminated soil and surface and recreational waters, have also been documented as sources (Bhandari et al., 2015; Chacín-Bonilla, 2017; Li et al., 2020). In addition, there is some evidence that ingestion of contaminated water may be a source of exposure. In a study from Venezuela, cyclosporiasis was strongly associated with areas of extreme poverty, poor sanitation and contact with faecally contaminated soil (Chacín-Bonilla et al., 2007).

Person-to-person transmission is virtually impossible because oocysts excreted in faeces require a maturation period in the environment in order to sporulate and become infective. Attempts at experimental infection of humans with oocysts have not been successful, suggesting the need for environmental triggers to initiate infection, which are not yet fully understood (Alfano-Sobsey et al., 2004).

Humans are the only confirmed reservoir for *C. cayetanensis*, although a small number of studies have isolated oocysts resembling those of *Cyclospora* or detected *Cyclospora* by molecular methods in animal faeces (Chu et al., 2004; Ortega & Sanchez, 2010). No evidence of tissue infection was identified in these studies. Oocysts are typically shed in relatively low numbers (Strausbaugh & Herwaldt, 2000), but other studies suggest shedding can reach 10⁴ oocysts/gram (Giangaspero et al., 2015b). Shedding can be intermittent and can also occur when the host is asymptomatic or for many weeks after symptoms resolve in untreated patients (Shields & Olson, 2003).

The relative infectivity of *C. cayetanensis* is unknown, but based on the high infectivity of oocysts of similar enteric protozoa (e.g. *Cryptosporidium*), it is probably high also (Chacín-

Bonilla, 2017).¹ The incubation period for cyclosporiasis is approximately 1 week from the point of ingestion (Adams & Ortega, 1999), but has ranged from 2 to 14 days in reported outbreaks (Heymann, 2015). The likelihood of diarrhoeal symptoms decreases with each subsequent infection (Ortega et al., 1997), suggesting a build-up of immunity from disease, but not from infection. However, outbreaks have been reported in endemic areas which could suggest that acquired immunity may not be long lasting or that geographic distribution of *C. cayetanensis* is variable (Almeria et al., 2019).

5 Detection methods

Cyclospora oocysts can be detected in environmental samples using microscopy or molecular methods, but standard methods are not available (Chacín-Bonilla, 2017; Totton et al., 2021), and both approaches have limitations. Detection of *Cyclospora* via microscopy is complex, resource-intensive and may not differentiate between species and viable and non-viable oocysts. Molecular methods cannot determine viability of oocysts and are not widely used.

6 Source and occurrence

Limited numbers of studies have detected *C. cayetanensis* oocysts in wastewater, surface water, groundwater, drinking-water and on crops. However, studies quantifying viable oocysts in the environment are limited (Chacín-Bonilla, 2017). Little is known about the effects of environmental conditions on the rate of sporulation and on the viability of oocysts, but they are generally hardy and environmentally resistant (Chacín-Bonilla, 2017). Sporulation in faeces usually takes at least 7 days, with about 40% of oocysts sporulating within 14 days at temperatures between 23 °C and 32 °C (Ortega et al., 1994; Chacín-Bonilla, 2017).

6.1 Faecal pollution sources

Oocysts have been detected in raw sewage in the small number of studies conducted in low-income settings, but there are no studies reporting concentration or survival in sewage. No data are available for the presence or survival of *C. cayetanensis* in sludge (Chacín-Bonilla, 2017). Oocysts have been detected in wastewater in high- and middle-income settings, but the number of studies conducted is small. Detection rates have varied from 13% in treated wastewater plant effluent in southern Italy to 73% in wastewater from a primary oxidation lagoon in Peru (Chacín-Bonilla, 2017). However, no data were available on the concentration or survival of *C. cayetanensis* in wastewater.

6.2 Surface water

Oocysts have been detected in river water in a small number of studies conducted in low- and high-income settings, with detection rates varying from 0.2% to 63% of samples (Chacín-Bonilla, 2017). The estimated concentration of the parasite in Guatemalan rivers was equal to

¹ From experiments with human volunteers, from epidemiological evidence and from experimental animal studies. Based on the criteria established by WHO in the *Guidelines for drinking-water quality* as follows: High means infective doses can be 1–10² organisms or particles, Moderate 10²–10⁴ and Low > 10⁴.

or greater than 1500 oocysts/L (Giangaspero et al., 2015b) and an irrigation canal in Egypt contained 1900 oocysts/L (Bern et al., 1999).

Cyclospora can survive for months in surface waters, depending on the temperature (Chacín-Bonilla, 2017).² Sporulation rates are often used as a proxy measure for persistence. Oocysts have been shown to survive for 2 months in water at 4 °C without sporulation but then sporulate within days of being transferred to water at 30 °C (Ortega et al., 1998). At 37 °C, only 5% of oocysts sporulated within 2 months, equivalent to a 1.3 log₁₀ reduction in infectivity.

6.3 Groundwater

Very limited information is available on the presence of *Cyclospora* in groundwater. A molecular survey of *Cyclospora* in environmental samples conducted in southern Italy from 2012 to 2014 detected the pathogen in 6.2% of well water samples (Chacín-Bonilla, 2017). Oocysts were isolated from shallow groundwater in Egypt in concentrations of 700 oocysts/L but not in deep groundwater (El-Karamany et al., 2005). The results suggested that sewage contamination was an issue.

6.4 Drinking-water

Cyclospora have been detected in public drinking-water supplies and in drinking-water packaged in bags (sachets) in a small number of studies in low- and high-income settings (Chacín-Bonilla, 2017). In Italy, 30% of tap water samples had detectable *Cyclospora* with concentrations ranging from 4 to 11 oocysts/L (Giangaspero et al., 2015a).

6.5 Soil and crops

Studies have detected *Cyclospora* on fruit, vegetables and herbs grown in low- and high-income settings and in shellfish, but no data are available on concentrations (Chacín-Bonilla, 2017). A molecular survey of *C. cayetanensis* in environmental samples conducted in southern Italy from 2012 to 2014 detected the pathogen in 12% of soils and 12% of vegetables (Giangaspero et al., 2015b). More recently, Li et al. (2020) reviewed the detection of *C. cayetanensis* on a variety raw fruits and vegetables in ten countries. The average prevalence was 3.9% and ranged from 1.4% to 67%. The occurrence of outbreaks of cyclosporiasis associated with fresh produce between 1995 and 2019 has been summarized by Hadjilouka et al. (2020). The type of produce as a source of outbreaks was basil (34%), raspberries (31%), cilantro (10%), lettuce (10%), salad mix (9%) and others (6%). Freezing or heating to high temperatures appears to inactivate oocysts in food products (Smith et al., 1997; Sathyanarayanan & Ortega, 2006).

6.6 Surfaces

No data are available for the presence of *Cyclospora* on surfaces (Chacín-Bonilla, 2017), but oocysts are very sensitive to desiccation where the oocysts wall ruptures after 15 min (Long et al., 1991).

² Based on the criteria established by WHO in the *Guidelines for drinking-water quality* as follows: Detection period for infective stage in water at 20 °C: Short, up to 1 week; Moderate, 1 week to 1 month; Long, over 1 month.

7 Significance of sanitation

Safe sanitation systems are an important barrier against the spread of *C. cayetanensis*. The pathogen is transmitted by the consumption of contaminated foods and water and ingestion or contact with soils. Since humans are the only known reservoir for *C. cayetanensis*, a sanitation service chain that separates human faeces from the environment and human contact should effectively control transmission of *C. cayetanensis* (WHO, 2022). Adequate management of faecal wastes reduces the risk of contamination of crops through irrigation or fertilization with treated wastewater or biosolids.

8 Significance in drinking-water

There is moderate evidence that *C. cayetanensis* has caused outbreaks in drinking-water, but not on the same scale as other protozoan pathogens. An outbreak of cyclosporiasis was reported from Nepal in 1994, where drinking-water consisting of a mixture of river and municipal water was associated with infections in 12 of 14 foreign soldiers (Rabold et al., 1994). In 1995, the first documented outbreak of cyclosporiasis in the United States of America was associated with contaminated tap water in a hospital setting. However, oocysts were not detected in water samples and a follow-up study proposed that the outbreak may have been foodborne (Huang et al., 1995; Ortega & Sanchez, 2010). In Haiti, the use of a particular well was the only risk factor associated with *Cyclospora* infection in a rural community in 2001 (Lopez et al., 2003). Between 2004 and 2010, at least 199 waterborne protozoan outbreaks were reported worldwide, of which only three (1.5%) were attributed to *C. cayetanensis*. Only one, a mixed outbreak with *Cryptosporidium*, was attributed to drinking-water (Baldursson & Karanis, 2011). In 2008 in Puerto Rico, an investigation of recurrent diarrhoea in a rural community identified a faulty drinking-water supply as a probable source of *Cyclospora* infections (Brunkard et al., 2011). In 2013, an outbreak of cyclosporiasis which involved three Polish travellers to Indonesia, was linked to the consumption of contaminated drinking-water (Bednarska et al., 2015). However, no local investigation of the drinking-water was conducted. *Cyclospora* were identified in a drinking-water storage tank during the outbreak but not detected in a sample from this well. Similarly, drinking-water has been reported as a risk factor for other clusters of cases or outbreaks without corroboration from further investigations (Shlim et al., 1991; Hoge et al., 1993; Bern et al., 1999; Tandukar et al., 2013; Bhandari et al., 2015; Ma et al., 2022).

9 Prevention and management

Systematic approaches that identify and control faecal contamination along the whole sanitation service and drinking-water supply chains are needed to manage microbial risks, including from *C. cayetanensis*.

The effectiveness of sanitation and drinking-water treatment systems depends on the selection and operation of technologies and the characteristics of the target organisms. Treatment processes based on physical removal (e.g. size exclusion by filtration through media or soil in septic tank drainage areas) or settling in wastewater treatment processes will reduce concentrations of protozoa such as *C. cayetanensis*. Oxidizing chemical disinfectants, such as

chlorine, are likely to have limited effectiveness against *C. cayetanensis* at the doses applied to drinking-water.

9.1 Sanitation systems

Sanitation safety plans should include control measures to safely collect, treat, discharge or reuse wastewater and faecal sludges (WHO, 2022). There are little data on removal of *Cyclospora* by wastewater treatment, but it will likely be similar to the removal of *Cryptosporidium*.

Studies have reported that wastewater treatment based on physicochemical and biological treatment with activated sludge partially removes *Cyclospora*, but treated effluent may remain contaminated with viable oocysts (Chacín-Bonilla, 2017). A study in Arizona detected *Cyclospora* by molecular methods in 25% of influent to a wastewater treatment plant and in 13% of effluent samples (Kitajima et al., 2014). In Italy, 21% of treated wastewater samples contained *Cyclospora* by molecular methods (Giangaspero et al., 2015b). There was a lower prevalence of *Cyclospora* (13%) in effluent from a plant with tertiary treatment, including membrane filtration. This study also noted that soils treated with wastewater effluents were more likely to be contaminated with *Cyclospora* than untreated soils (Giangaspero et al., 2015b). Since oocysts are sensitive to high temperatures and low-moisture conditions, *Cyclospora* are not expected to survive anaerobic digestion of biosolids, although evidence is lacking to support this theory (Gerba et al., 2002).

9.2 Drinking-water supplies

Within a water safety plan (WHO, 2012; 2023), control measures that can be applied to manage potential risks from *C. cayetanensis* include protection of source waters from contamination by human faecal waste, application of adequate treatment processes and protection of drinking-water quality during storage and distribution.

There are no data on removal of *Cyclospora* by water treatment. However, filtration processes that are effective against *Cryptosporidium* should be effective at removing *Cyclospora*, which are slightly larger (8–10 µm compared to 4–6 µm). An outbreak of cyclosporiasis linked to a chlorinated but unfiltered drinking-water system in Nepal supports this opinion (Rabold et al., 1994). Based on their cell wall structure, *Cyclospora* oocysts are probably highly resistant to chlorine³ and are expected to be as sensitive to UV light as other protozoan cysts and oocysts (Linden & Murphy, 2017).

Due to the long environmental persistence of *Cyclospora* oocysts and their probable high resistance to chemical disinfectants, the absence of *E. coli* (or alternatively, thermotolerant coliforms) is not a suitable indicator of the absence of *C. cayetanensis* in drinking-water.

³ Within pathogen species and groups, there are likely to be variations in resistance, which could be further impacted by characteristics of the water supply and operating conditions. Resistance is based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: 99% inactivation at 20 °C where, generally, Low represents a Ct₉₉ of < 1 min.mg/L, Moderate 1–30 min.mg/L and High > 30 min.mg/L (where C = the concentration of free chlorine in mg/L and t = contact time in minutes) under the following conditions: the infective stage is freely suspended in water treated at conventional doses and contact times, and the pH is between 7 and 8.

However, the presence of *E. coli* is a suitable indicator of the potential presence of *C. cayetanensis*.

10 ICD-11 classification

XN7VL	<i>Cyclospora</i>
XN4BR	<i>Cyclospora cayetanensis</i>
1A3Y	Other specified protozoal intestinal infections cyclosporiasis

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Annex. Approach to content development and declarations of interest

The background documents were developed drawing from the following resources (i) microbial fact sheets from the fourth edition of the WHO Guidelines for Drinking-Water Quality (ii) pathogen-specific chapters in the Global Water Pathogen Project (GWPP) book “*Sanitation and Disease in the 21st Century*”, an update of the seminal book by Feachem et al. 1983 “*Sanitation and Disease: Health Aspects of Excreta and Wastewater Management*” and (iii) a rapid review of the literature at the start of the development of the background documents and before finalization in December 2023, focusing on identifying additional recent high quality literature reviews on a given pathogen. The documents were prepared by members of the WHO drinking-water quality microbiology group as well as external scientists with expertise in the relevant pathogens. The documents went through several iterations based on inputs from members of the WHO drinking-water quality microbiology working group and WHO technical staff. In addition, each document was reviewed by at least two external peer reviewers including authors of the GWPP book chapters. The extensive review process was designed to ensure technical accuracy and to identify any data gaps. The background documents were discussed at three expert meetings to resolve any issues and ensure the content reflected a consensus of expert opinions. Upon satisfactory resolution of any further data gaps or issues, the background documents went through technical editing before finalization.

Before the key expert meetings mentioned above, and for this work more broadly, all microbiology working group members submitted declarations of interest to WHO. These were to disclose any potential conflicts of interest that might affect, or might reasonably be perceived to affect, their objectivity and independence in relation to the subject matter of the guidance. WHO reviewed each of the declarations and concluded that none could give rise to a potential or reasonably perceived conflict of interest related to the subjects discussed at the meetings or covered by the guidance.